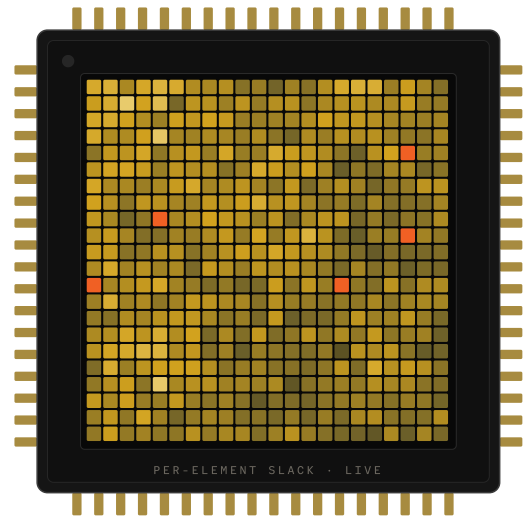


TIMING HEALTH FOR FPGA FLEETS

Run every FPGA at its measured limits.

Vendor timing models are built for the slowest die, at the hottest corner, at the end of its life. Fluid Silicon measures each of your chips in production, at speed and with no downtime, so you can reclaim margin, see failure coming and retire hardware on evidence.



Illustrative die. Each cell stands for one logic element's measured slack.

30–54%* timing margin a chip carries today to cover the worst-case corner	< 20 ps per-link measurement precision, error plus drift	≈ 1s to measure every logic element on the chip
< 1% of LUTs and under 5% of flip-flops on a production shell	< 2% delay impact on your design	0 downtime while monitoring runs

*Fluid Silicon measurements. Varies with variation pattern, workload, vendor and device age.

IN THIS BRIEF

- 01 The problem
- 02 The platform
- 03 Lifecycle and adoption
- 04 Security and integration
- 05 Devices and next steps

Margins are set once, for a chip that is statistically rare.

Vendor timing models cover the worst die: highly defective, at elevated voltage, at high temperature and at the end of its life. Most of your chips are nowhere near that corner, yet every one of them runs as if it were.

VARIATION

Variation you only see at scale

At 16, 14 and 7 nm, defects show up as more than 20% random variation on top of systematic variation. A fleet-wide margin has to cover the worst chip.

AGING

Aging nobody measured

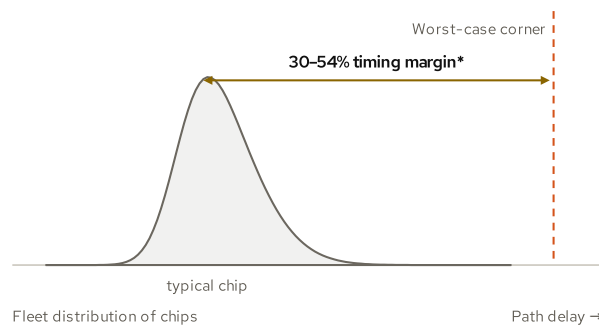
Timing drifts with time, temperature and voltage. Cards now stay in service six years and longer, in hotter racks than anyone planned for.

VISIBILITY

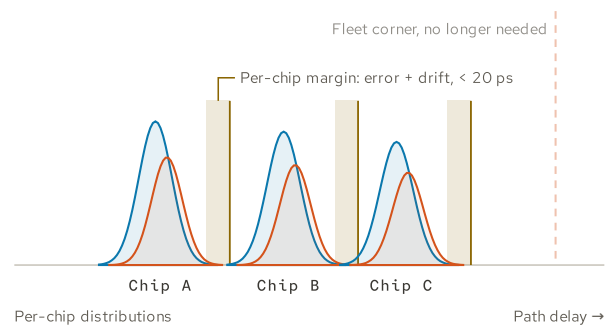
Silicon as a black box

When a card fails, nobody can say which logic caused it. There is no health signal to plan around and no evidence to attribute errors.

Today: one margin for the fleet



Measured: a margin per chip



— Fleet of chips — LUT links — Carry links — Per-chip sign-off

Schematic, not to scale. Measured per chip, the margin each chip needs shrinks to measurement error plus drift until the next measurement. *Fluid Silicon measurements.

FPGA fleets are now infrastructure.

>1M

Azure hosts with FPGA-based SmartNICs, deployed on all new Azure servers since late 2015.

Firestone et al., "Azure Accelerated Networking: SmartNICs in the Public Cloud," NSDI 2018.

<https://www.usenix.org/conference/nsdi18/presentation/firestone>

"Currently, Azure has 5 generations of FPGA boards actively in deployment."

Rydberg et al., "Hyperscale FPGA Engineering Systems at Microsoft," FPGA 2026. <https://doi.org/10.1145/3748173.3779203>. The same paper describes supporting multiple board generations for a decade or longer. Cited for context; Fluid Silicon is not affiliated with Microsoft.

A timing-health layer for FPGA fleets.

Fluid Silicon gives each FPGA fine-grained timing introspection. Chips measure the health of their own logic at speed, tune voltage and frequency to what they can sustain, and route around resources that degrade. Monitoring and modeling are read-only. Tuning and repair are separate, opt-in steps.

MONITOR

Per-element timing, at operating speed

Timing slack on individual logic elements, including LUTs, carry blocks and the links between them. A full sweep takes about one second, with each link resolved to under 20 ps.

MODEL

An aging curve for every card

Each element is tracked for the life of the card, its aging fitted to its own temperature history, with a predicted time-to-threshold. Outliers are flagged early.

TUNE

Voltage and frequency, per chip

Adaptive frequency scaling finds the Fmax a chip can sustain for long periods. Adaptive voltage scaling trims voltage for better compute per watt. Both can be coordinated per device, per server or across the fleet.

REPAIR

Swap out what is degrading

Defective or degrading resources are replaced with healthier, faster ones, inside the maintenance windows you schedule.

FLUID SILICON / THE HEALTH LOOP

BEFORE DEPLOYMENT

Characterize and map

LUTs and carry blocks measured at speed.
Weak resources mapped around.

IN PRODUCTION

Monitor

Every logic element swept in about one second while your design keeps running.

Model

An aging curve per card, per element. Predicts time-to-threshold.

Tune and repair

Adaptive frequency scaling
Finds the Fmax the chip can sustain.

Adaptive voltage scaling
Better compute per watt.

Repair
Degrading resources swapped for healthier, faster ones.

AFTER DEPLOYMENT

Characterize

Cards leaving service measured at speed, with the evidence to resell, reuse or retire them.

VENDOR-NEUTRAL · DESIGN-AGNOSTIC · ZERO DOWNTIME · OPT-IN ACTIONS

The health loop runs for the life of each card: characterization before deployment, a monitoring, modelling and tuning loop in production, and characterization after service. Every measurement tightens the next margin.

Across the whole life of a card.

BEFORE DEPLOYMENT

Characterize and map

Measure every LUT and carry block at speed to expose weak resources, then keep critical paths away from them.

IN PRODUCTION

Monitor, model, compensate

Watch timing online while the design runs, model how each chip changes with time and temperature, and adapt as it does.

AFTER DEPLOYMENT

Decide what happens next

Characterize cards leaving service so you can resell, reuse or retire them, with data to support the decision.

One health metric, on every vendor.

Measured slack as a share of the clock period, per logic-element link. The same number means the same thing on an AMD part and an Altera part, so fleet dashboards, alert thresholds and aging models work across both.

MEASURE	VALUE
Margin a chip carries today for the worst-case corner	30–54%*
Margin needed with per-chip measurement	Measurement error plus drift
Per-link precision (error plus drift)	< 20 ps
Time to measure every logic element	≈ 1 s
LUT and flip-flop overhead on a production shell	< 1% LUTs · < 5% FFs
Delay impact on your design	< 2%
Downtime to monitor	None

*Fluid Silicon measurements; they vary with variation pattern, workload, vendor and device age.

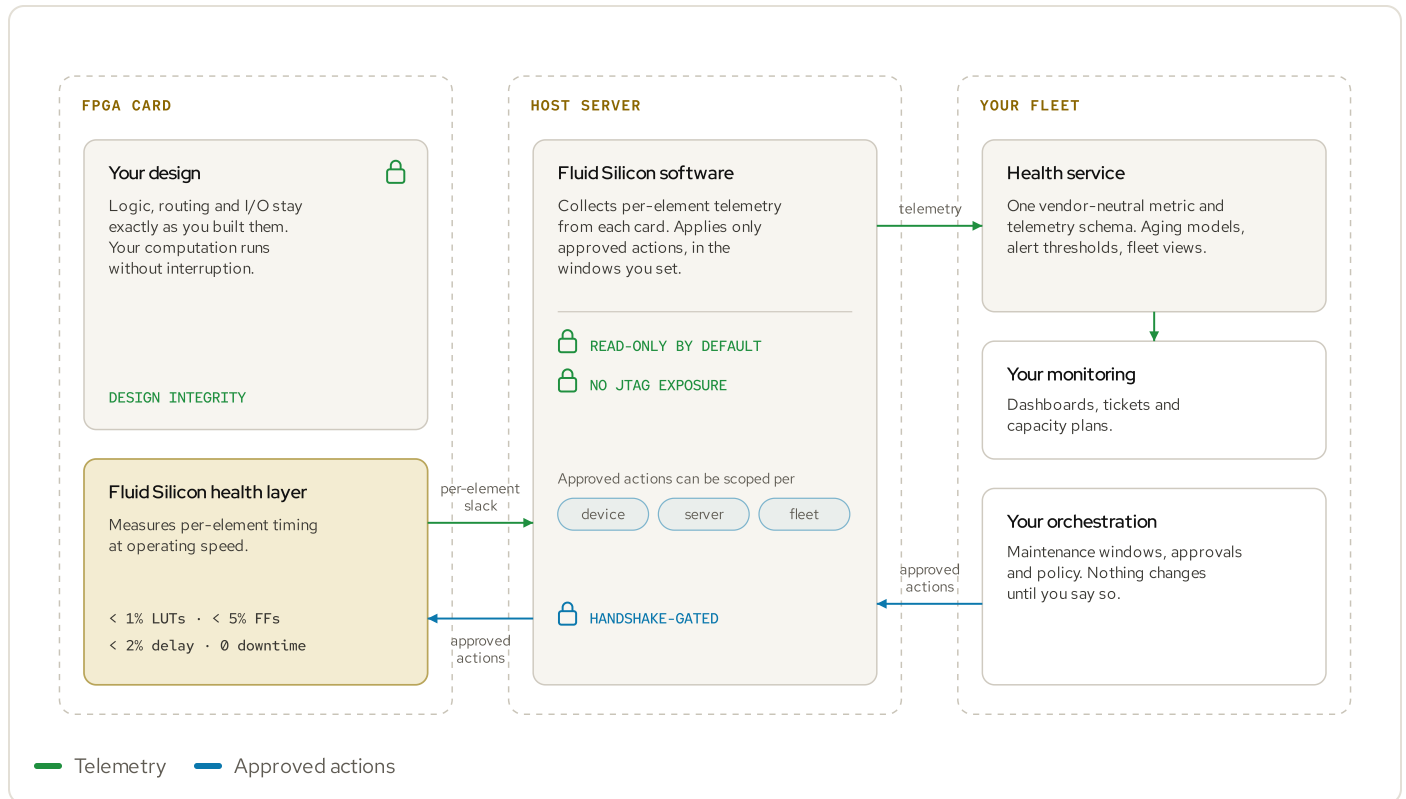
Adopt it one step at a time.

Every step is opt-in, and you can stay at any step as long as you want.

01	02	03	04
Evaluate	Observe	Optimize	Repair
A sample of cards in a staging environment, with a per-card timing and variation report.	Monitoring and aging models in production. Operating points do not change.	Voltage and frequency tuning per policy, in the windows you schedule.	Degrading resources swapped for healthy ones, through your change process.
Read-only	Read-only	Opt-in · scheduled · handshake-gated	Opt-in · scheduled · handshake-gated

Built to run beside production designs.

Telemetry flows out from each card to your fleet. Actions flow back only after you approve them, and only through a handshake.



INTEGRITY

Your design stays intact

Logic, routing and I/O are left as you built them. The health layer occupies a small, fixed region beside them.

ACCESS

No JTAG exposure

No debug port has to be opened on production systems for monitoring to work.

AVAILABILITY

Zero downtime

Your design keeps computing for the whole measurement. A full sweep takes about a second.

FOOTPRINT

Small and design-agnostic

Under 1% of LUTs and 5% of flip-flops on a production shell, with under 2% added delay.

CONTROL

Deterministic, scheduled changes

Voltage, frequency and repair actions are opt-in, scheduled and handshake-gated.

DATA

One vendor-neutral view

The same health metric, telemetry schema, alert thresholds and aging model on AMD and Altera parts.

Supported devices.

One platform across AMD and Altera families, with the vendor-specific integration handled by Fluid Silicon.

VENDOR	FAMILY	PROCESS NODE	STATUS	NOTES
AMD	7-series	28 nm	Supported	
AMD	UltraScale	20 nm	Supported	
AMD	UltraScale+	16 nm	Supported	Includes Virtex UltraScale+
Altera	Agilex 7	10 nm class	Select devices	Select devices, including F-Series
Altera	Stratix 10 / Arria 10	14 nm / 20 nm	In development	
AMD	Versal AI Core	7 nm	In development	

Status as of September 2026. RF-sampling adaptive SoCs for far-edge systems are on the roadmap.

Where it is used.

DATA CENTER AND HYPERSCALE

Multi-vendor fleets, years in service

Catch a failing chip before it costs capacity or sends errors downstream, trim voltage to cut power and raise frequency where a chip can hold it.

AEROSPACE AND DEFENSE

Far-edge systems on tight power budgets

Self-monitoring and repair from defects and degradation, aging prediction for mission planning, and adaptive voltage scaling where every watt counts. It complements radiation mitigation rather than replacing it.

Sensor architecture, deployment mechanics and calibration methods are shared with evaluation partners under NDA.

NEXT STEP

See per-element timing live on a supported device, then talk through an evaluation on your own cards.

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